11.9. A 2 MeV neutron collides elastically with an iron atom (^{56}Fe). What is the average temperature (corresponding to the maximum velocity) which can be ascribed to the iron nucleus after the collision?

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CHAPTER 12

The Origin of the Universe and Nucleosynthesis

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Today it is assumed that our universe consists of ~70% dark energy, ~25% dark matter, and ~5% of radiation and visible matter, see Table 12.1. Hydrogen, deuterium and most of the helium atoms in the normal matter part of the universe are believed to have been created 13.75 ± 0.11 billion years ago in a primary formation process referred to as the *Big Bang*, while all other elements have been formed — and are still being formed — in nuclear reactions in the stars and today also in our nuclear reactors. Most of these reaction processes can only be understood in an astrophysical context, as briefly outlined in this chapter, which also describes how nuclear science has provided much understanding about the universe, our solar system and our planets. The fusion reactions, which created the lightest elements, may in the future also provide us with a large source of energy in controlled thermonuclear reactors, see §19.17. It follows from the understanding of the processes involved that both fission and fusion used for power producing purposes should be named *nuclear power.*

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